

Iris Flower Classification

CodeAlpha Data Science Internship - Task 1

Author: Marwan Tamer

1. Project Overview

This project builds a machine learning classification model to predict the species of Iris flowers based on flower measurements. The model uses sepal length, sepal width, petal length, and petal width as input features.

2. Objective

- Load and explore the Iris dataset.
- Visualize feature relationships between flower species.
- Train a classification model using Scikit-learn.
- Evaluate model performance using accuracy, classification report, and confusion matrix.

3. Tools and Libraries

Tool / Library	Purpose
Python	Main programming language
Pandas	Data handling and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Advanced statistical visualizations
Scikit-learn	Dataset loading, model training, and evaluation

4. Dataset Description

The Iris dataset contains 150 records and 3 species: Setosa, Versicolor, and Virginica. Each species has 50 samples, which makes the dataset balanced. There are no missing values.

Column	Description
sepal length (cm)	Length of the flower sepal
sepal width (cm)	Width of the flower sepal
petal length (cm)	Length of the flower petal
petal width (cm)	Width of the flower petal
species	Encoded target class
species_name	Flower species name

5. Exploratory Data Analysis

The data was inspected using `head()`, `info()`, `describe()`, and `value_counts()`. The analysis showed that the dataset contains 150 complete records with balanced classes. Pairplot visualization showed that Setosa is clearly separated from the other two species, while Versicolor and Virginica are closer to each other. The correlation heatmap showed a strong positive relationship between petal length and petal width.

6. Model Building

The input features were separated from the target column. The dataset was split into training and testing sets using an 80/20 split. A Decision Tree Classifier was trained on the training data because it is simple, interpretable, and suitable for classification problems.

7. Evaluation Results

The model was evaluated using accuracy score, classification report, and confusion matrix. The model achieved 100% accuracy on the test dataset. The confusion matrix showed that all test samples were correctly classified.

Metric	Result
Accuracy	100%
Precision	1.00 for all classes
Recall	1.00 for all classes
F1-score	1.00 for all classes

8. Conclusion

The Decision Tree Classifier performed very well on the Iris dataset and achieved perfect accuracy on the testing set. This project demonstrates the basic machine learning workflow: data loading, exploration, visualization, training, prediction, and evaluation.